Lecture8: Clustering

STA 4724

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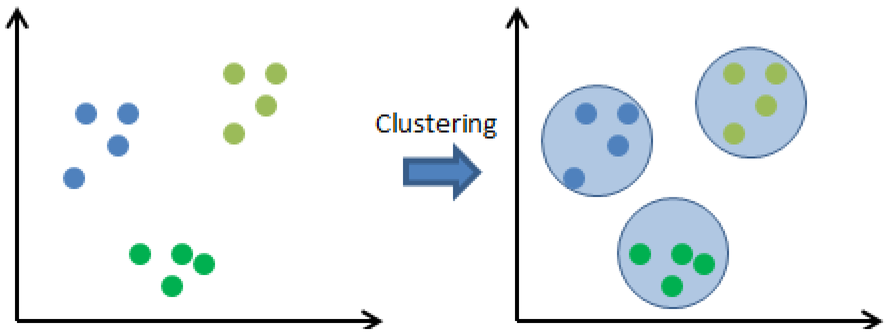
Clustering

- The process of dividing a given dataset into several groups (i.e., clusters)

- Data points in the same cluster are as similar as possible

- Data points in different clusters are dissimilar



- Clustering requires data, but not predefined labels
  - Unsupervised learning

- When the predefined label does exist, we should discuss *classification* instead (cover in later classes)

Clustering is Natural to Human

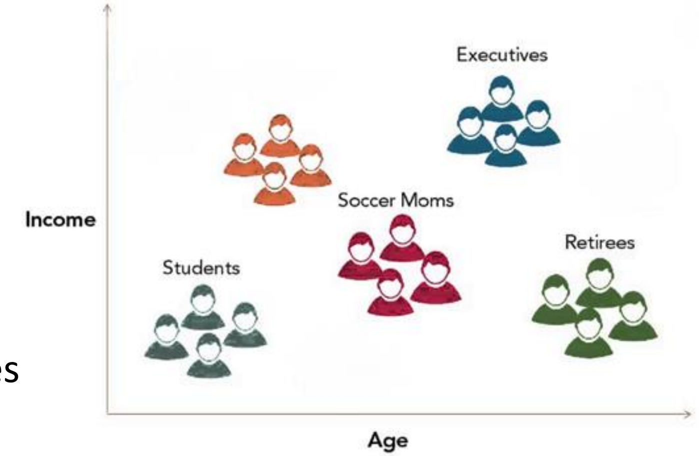
One corner in the store

- Imagine you are managing a supermarket
- Naturally, you put all the seafood in one section
- The meat section is usually not too far away
- TVs and video games are in another corner of the store

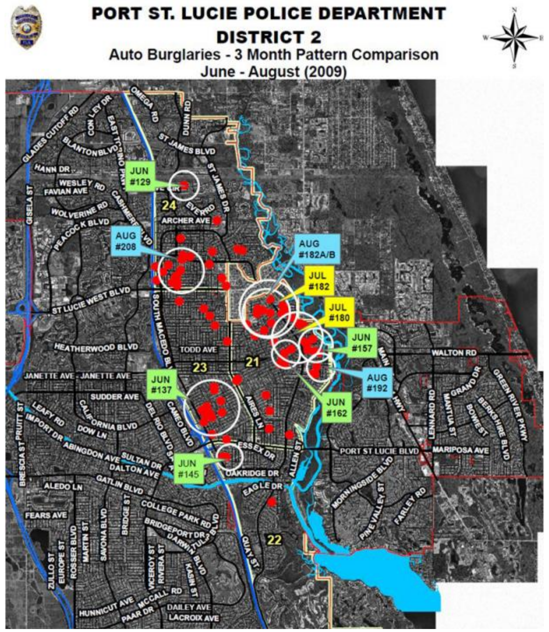
Another corner

Applications

- Many applications in computer science, market research, social science, biology, medical imaging …



- Two examples:
  - Customer segmentation
    - ❖ Group customs based on their backgrounds and shopping styles
    - ❖ The business owner can apply specialized marketing strategies



  - Crime analysis
    - ❖ Identify the "hot spots" where a similar crime has happened over a period of time
    - ❖ Help the law enforcement officials to manage their resources more effectively

Toward Supervised Learning

- Clustering is therefore a data exploration technique that contributes to our familiarity with the dataset

- Once suitable clusters have been created, we can provide them labels based on human knowledge

- These clusters can be the starting point to use supervised machine learning techniques

K-Means Clustering

- Clustering is NP-hard (non-deterministic polynomial-time hard)
  - Discussion on board
  - Finding the best solution is very computationally difficult

- K-Means is an efficient heuristic algorithm for clustering
  - Converges quickly to a local solution

- "K" in K-Means: the number of clusters

K-Means Clustering

- Iteratively assigns each point to one of the *K* clusters
  - Based on how close the point to the current means of the clusters

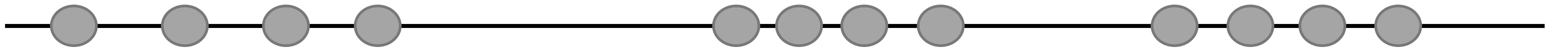
- A greedy algorithm
  - It is a heuristic algorithm that always takes the local optimal choices
  - the final solution depends on the initial conditions

- In general, a heuristic helps us solve a problem in a quick fashion by finding an approximate solution

Toy Example

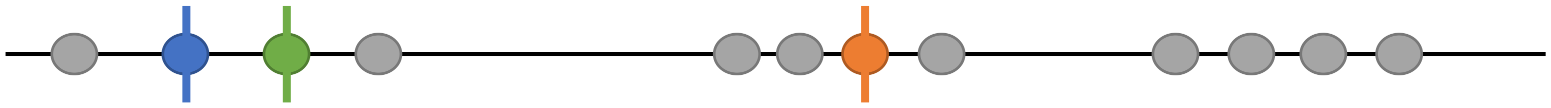
- Image we have some data points that we can plot on a line

- Visually, we know we want to put them into 3 clusters, i.e., K = 3



Toy Example: Step by Step

1. Initial 3 clusters by randomly selecting 3 distinct data points
   - They become the means of the clusters, for now



| denotes the mean of the **blue** cluster

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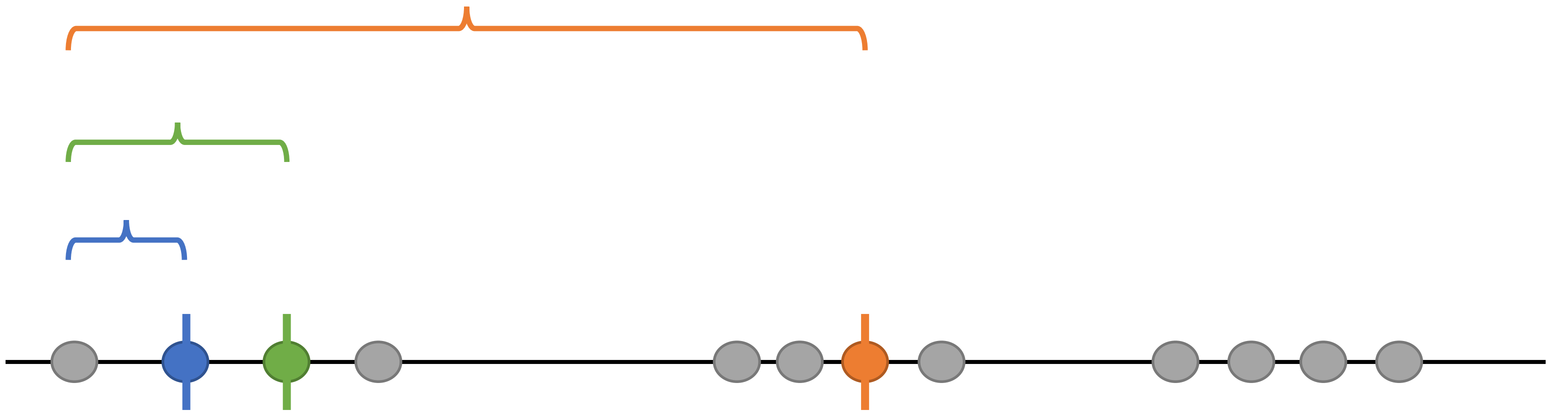

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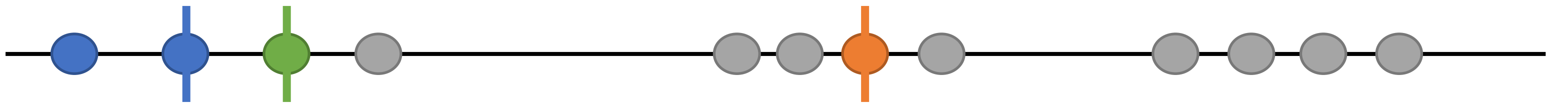

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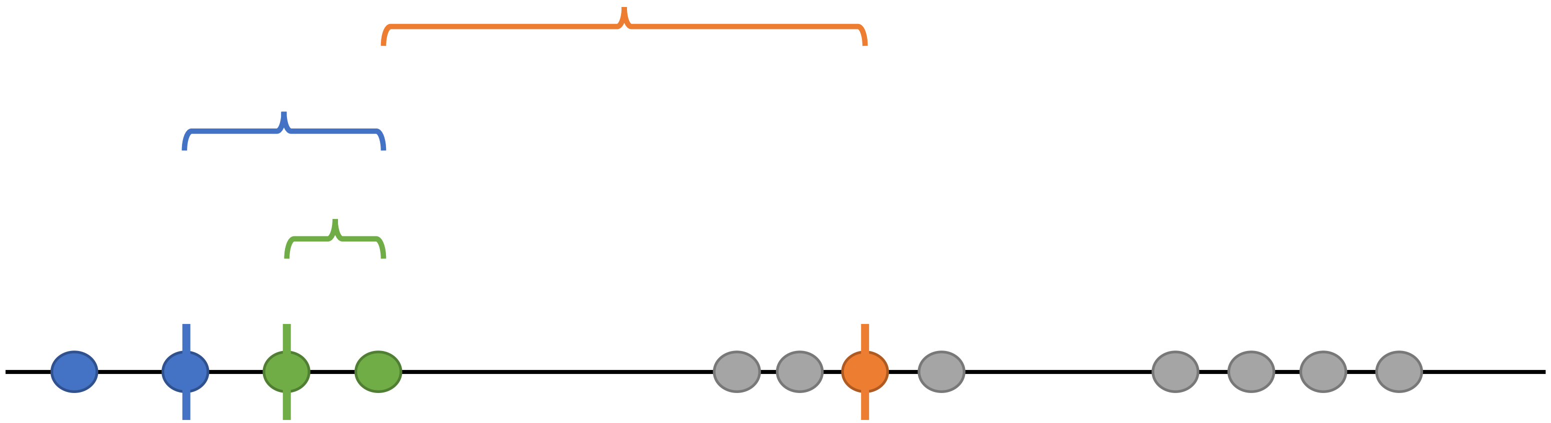

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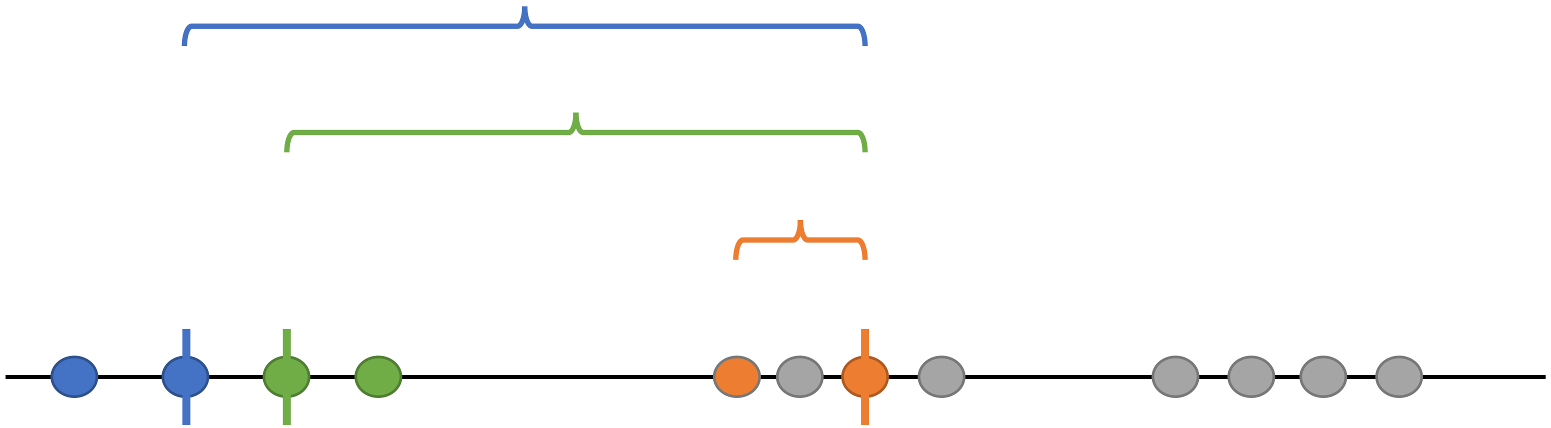

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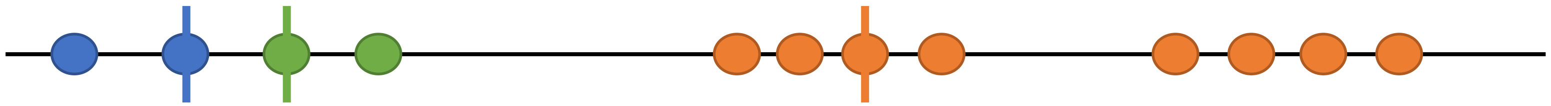

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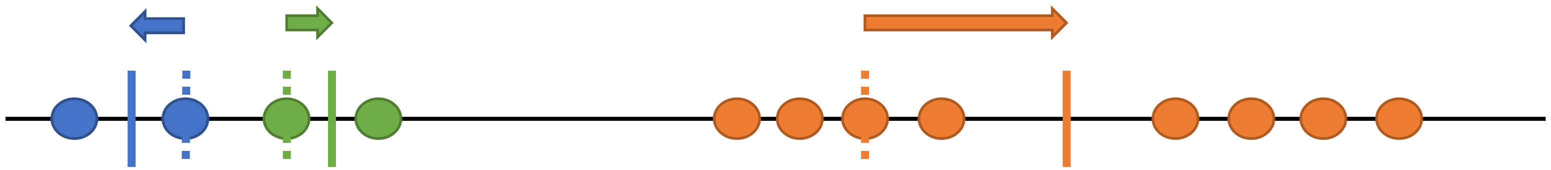

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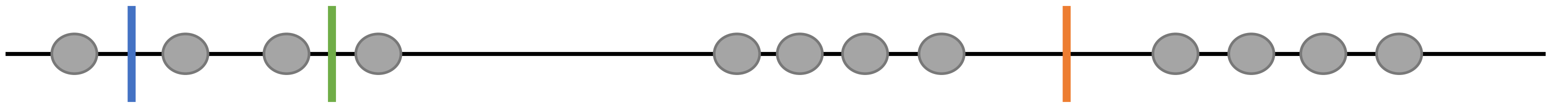
5. Reassign all the points again, using the **new** means



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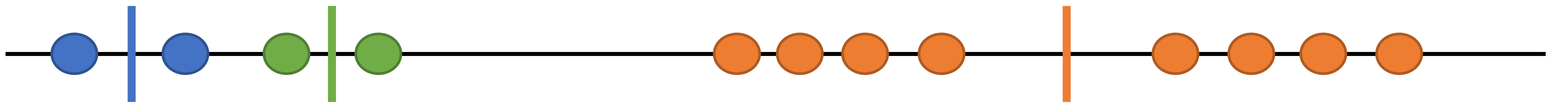

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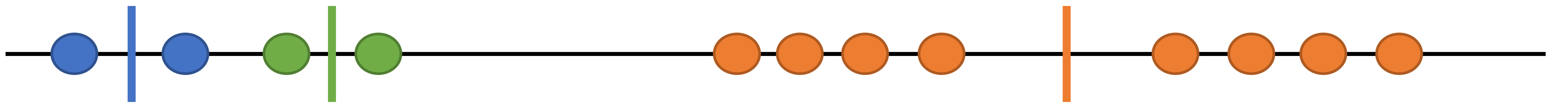
6. Repeat Steps 4-5 until the algorithm converges
   - convergence: no changes in the points assignment anymore



- So, we get a solution for this clustering problem

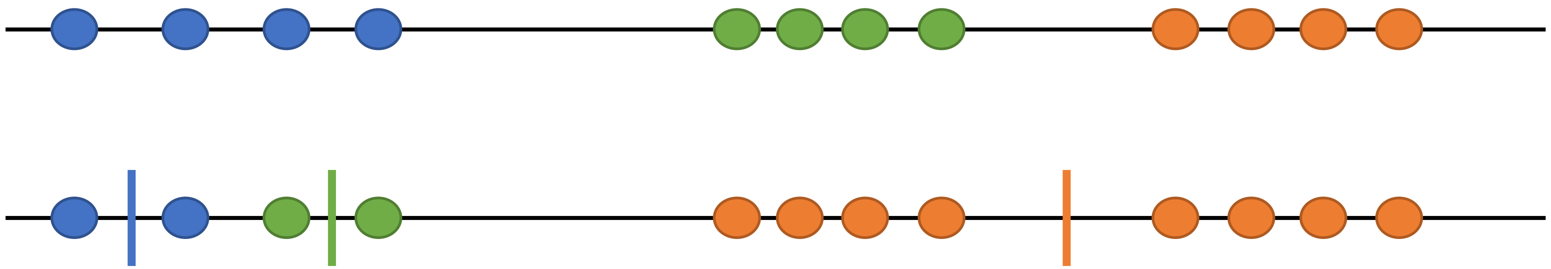
But…

- This solution is not matching the human solution

- Not a surprise --- Recall that *K*-Means does **not** guarantee convergence to the best solution

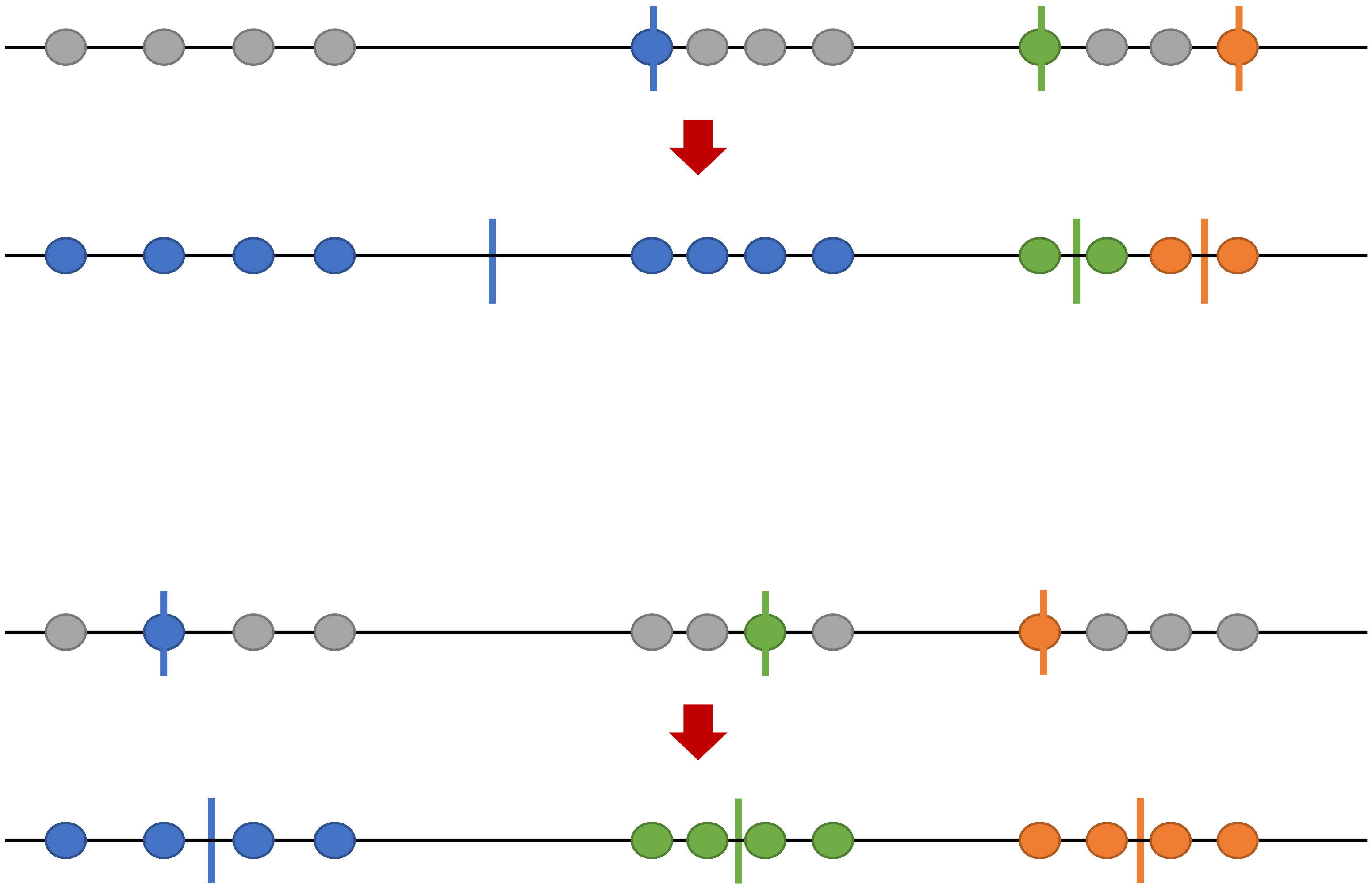
- Try *K*-Means with several different initializations, then pick the best result

Human solution →



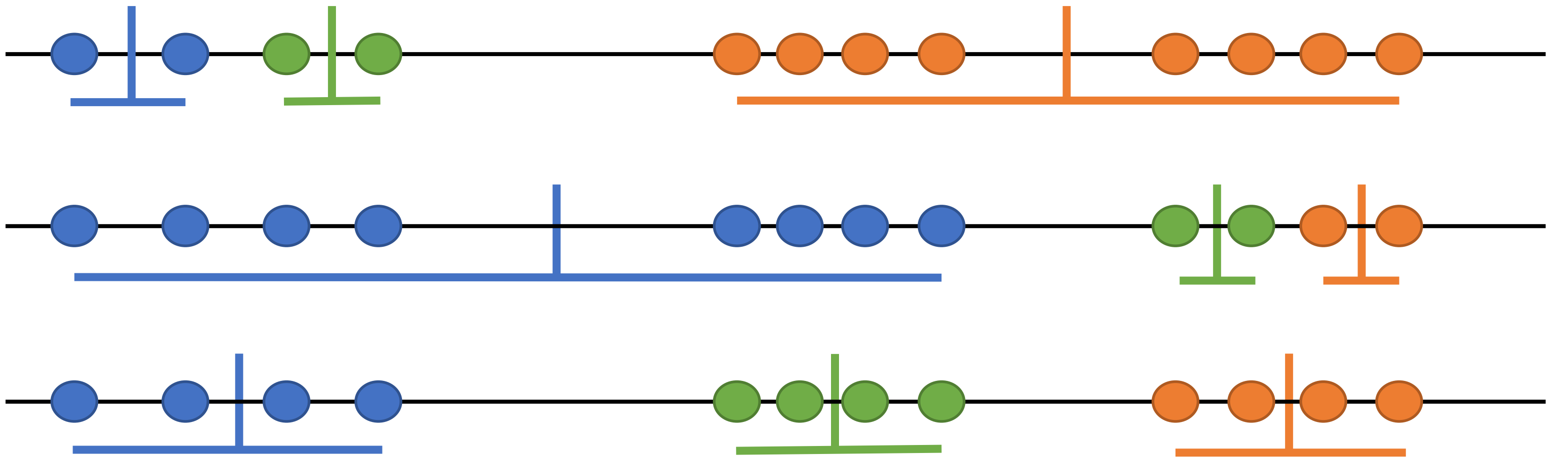
More solutions

- Different initializations may give different clustering solutions



Clustering Quality

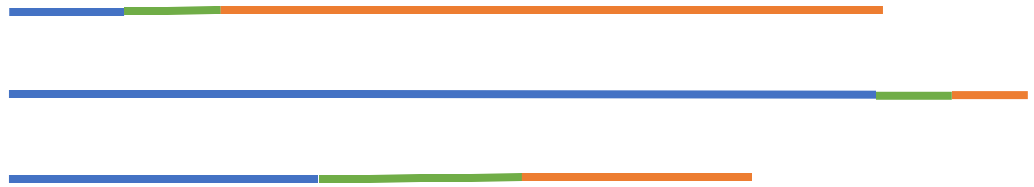
- Question: How to measure the quality of a solution?
- Minimize the sum of variation within each cluster (aka. *total intra-cluster variation*)



1st solution:

2nd solution:

3rd solution:



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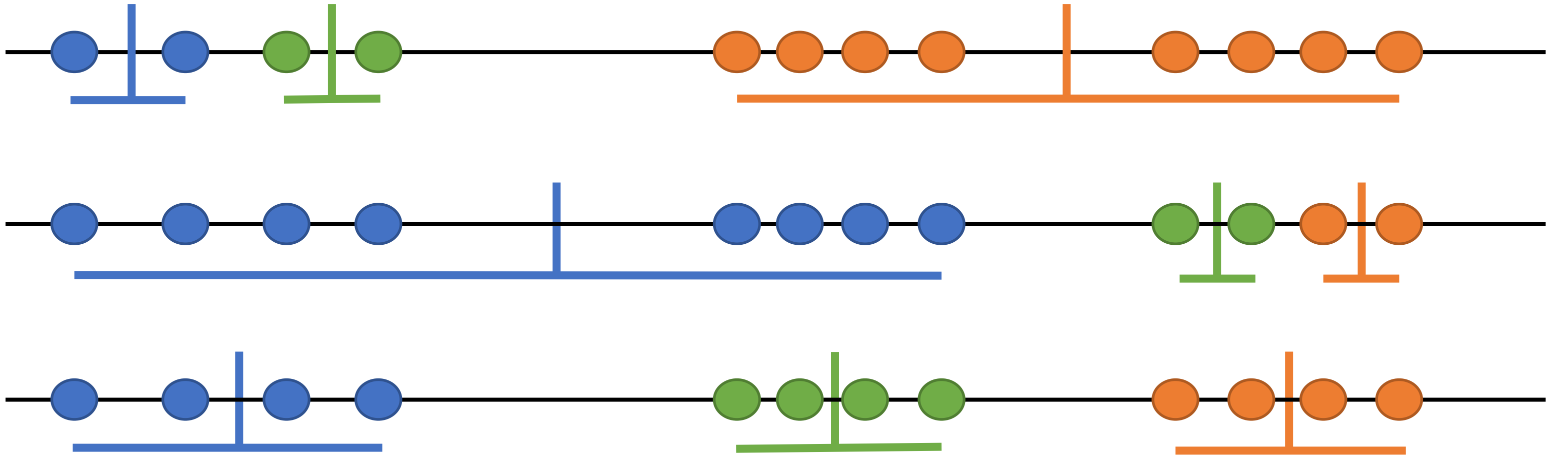

1st solution:

2nd solution:

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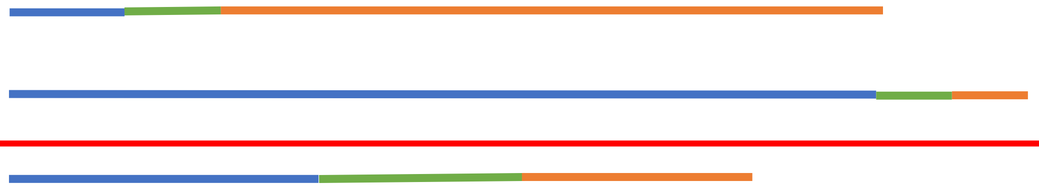


← The best solution so far

- Computer still doesn't know if this is the best solution overall, try more rounds!

One More Challenge

- In this example, by visualizing the data, it is obvious that we have 3 clusters

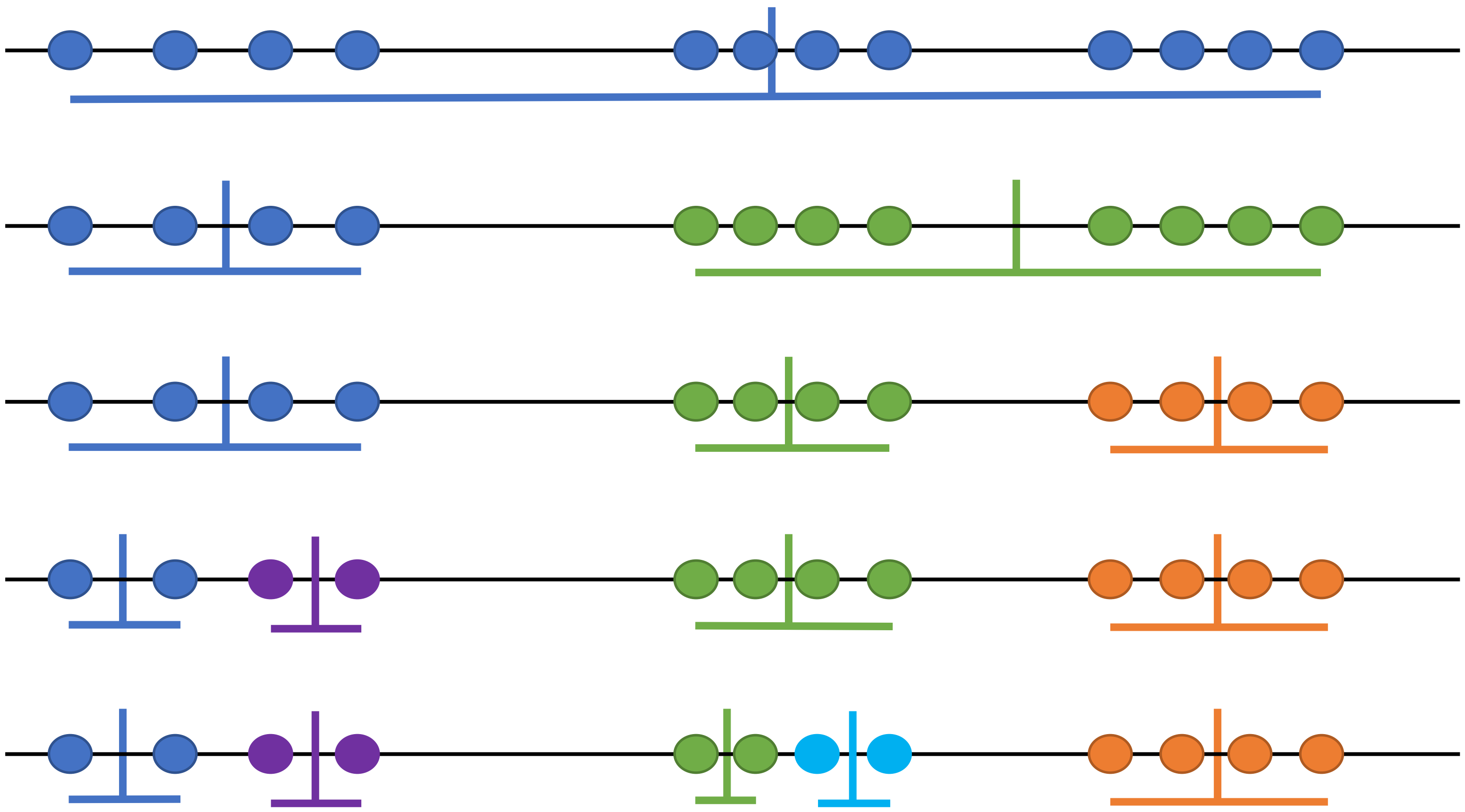
- Sometimes, it is not so clear how many clusters should be used
  - Especially, for high-dimensional dataset

- Question: How to mathematically find the best value for "*K*" (i.e., # of clusters)?

- Try different values for *K*, and keep track of the total intra-cluster variation

Find the Best Value for K

- $K = 1$
- $K = 2$
- $K = 3$
- $K = 4$
- $K = 5$



⋮

⋮

- $K = 12$ Each point is its own cluster

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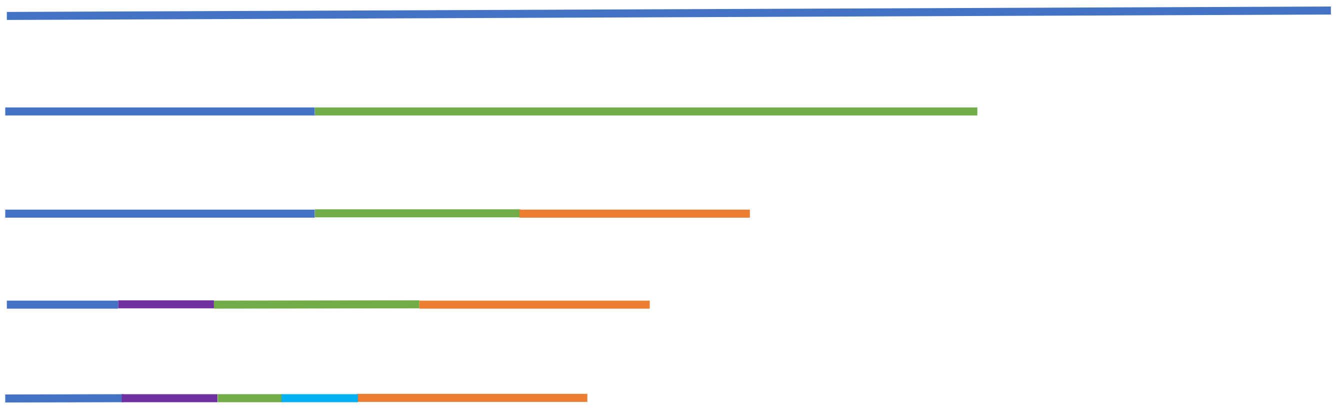


⋮

- Each time we add a new cluster, the total intra-cluster variation is getting smaller

⋮

- $K = 12$ Each point is its own cluster => Variation = 0

- However, $K = 12$ (# of the points) is obviously a **bad** choice for clustering
  - We are not getting any useful information

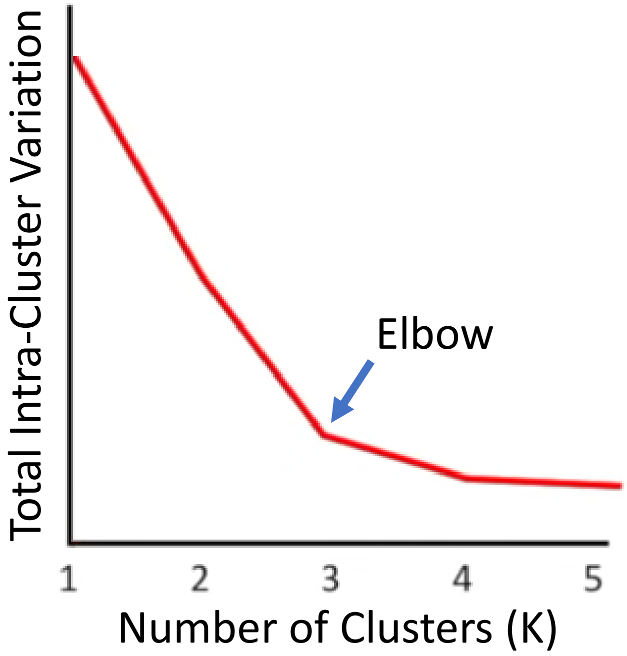
Elbow Method

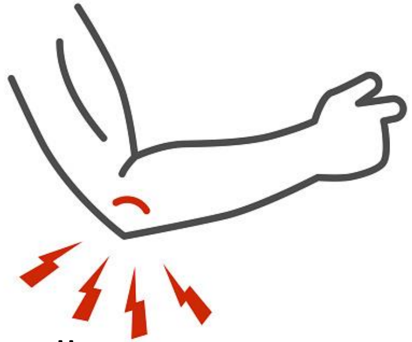
- Make a plot for the **total intra-cluster variation**

- Before K = 3, we see big reductions in variation. After that, the variation does not go down as quickly
  - K = 3 is called an "elbow" in this example

- Pick the best value for "K" by finding the "elbow" in the plot

- Finally, we have all the pieces for a complete K-Means algorithm





Elbow

More Complicated Examples

- Visualizing K-Means Clustering

High-Dimensional Data

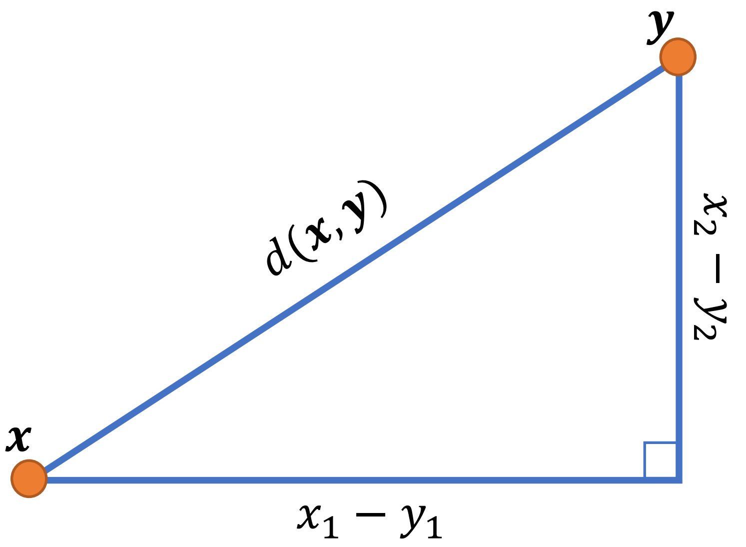
- While our toy example was on 1-D data, *K*-Means works for high-dimensional data

- The similarity between two high-dimensional points is measured by **Euclidean distance**

- Recall the definition from linear algebra class:
  - Given two n-D points: $\boldsymbol{x} = [x_1, x_2, \cdots, x_n]$ and $\boldsymbol{y} = [y_1, y_2, \cdots, y_n]$
  - Euclidean distance:

$$d(\boldsymbol{x}, \boldsymbol{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2}$$

  - Physically, this is the **length** of the **straight line** between $\boldsymbol{x}$ and $\boldsymbol{y}$



- Now, we can use *K*-Means for high-dimensional datasets

Time to Play

- Let’s see how to use *K*-Means in **python**!